

Targeting BRCA1 and BRCA2 Deficiencies with G-Quadruplex-Interacting Compounds

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G-quadruplex (G4)-forming genomic sequences, including telomeres, represent natural replication fork barriers. Stalled replication forks can be stabilized and restarted by homologous recombination (HR), which also repairs DNA double-strand breaks (DSBs) arising at collapsed forks. We have previously shown that HR facilitates telomere replication. Here, we demonstrate that the replication efficiency of guanine-rich (G-rich) telomeric repeats is decreased significantly in cells lacking HR. Treatment with the G4-stabilizing compound pyridostatin (PDS) increases telomere fragility in BRCA2-deficient cells, suggesting that G4 formation drives telomere instability. Remarkably, PDS reduces proliferation of HR-defective cells by inducing DSB accumulation, checkpoint activation, and deregulated G2/M progression and by enhancing the replication defect intrinsic to HR deficiency. PDS toxicity extends to HR-defective cells that have acquired olaparib resistance through loss of 53BP1 or REV7. Altogether, these results highlight the therapeutic potential of G4-stabilizing drugs to selectively eliminate HR-compromised cells and tumors, including those resistant to PARP inhibition.